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Next week

Tue
Wed
Thu
8-10 PM

Sat
Sun
Morning

Graphs

Prerequisites

- Arrays & 2D arrays (lists)
- Maps & Sets
- Heaps
- Queues & Stacks
- Recursion
- B. Trees (BFS & DFS)
- Oops

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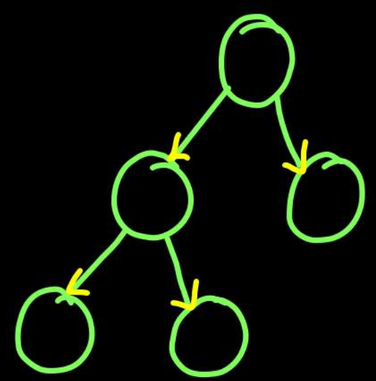
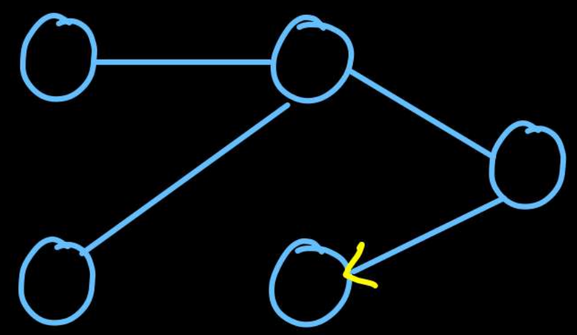
Nodes and Edges





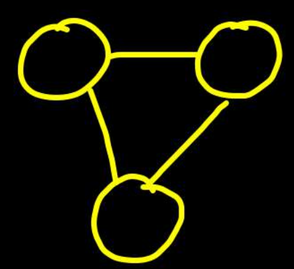
B.Tree

↓

One Way

No Cycles

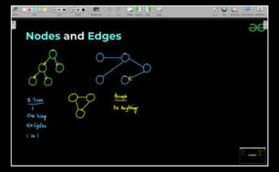
1 to 1



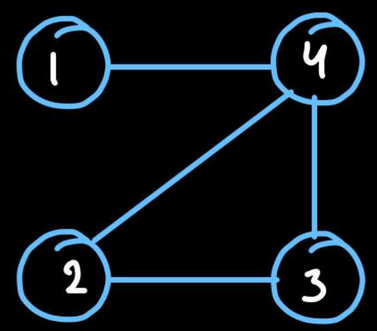
Graph

Do Anything

Edge Weight

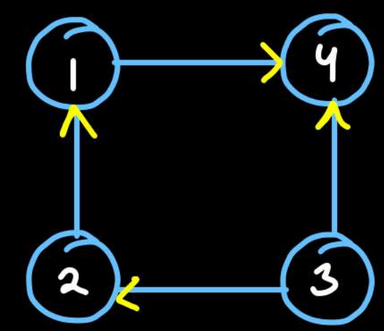


Directed and Undirected graph



undirected

cycle ✓



directed

no cycles

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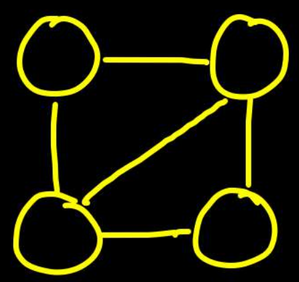
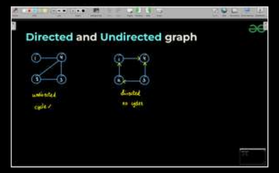
Cycle in a graph

Directed Graph → theda check karo

Undirected Graph → Any closed loop ✓

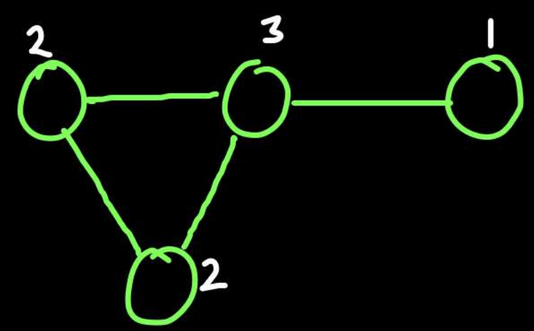


3 cycles

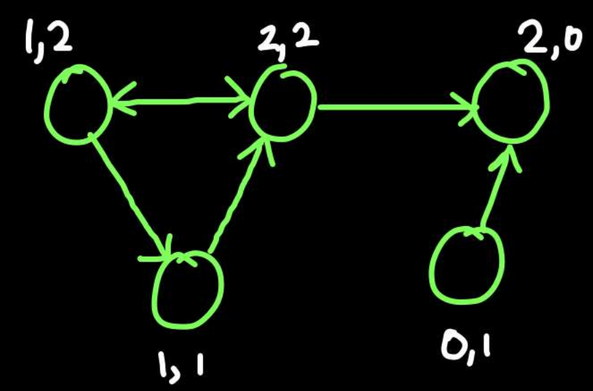


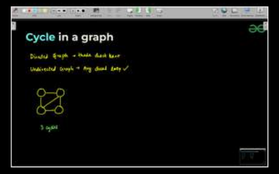
Degree of a Node

Undirected Graph



Directed Graph (Indegree & Outdegree)





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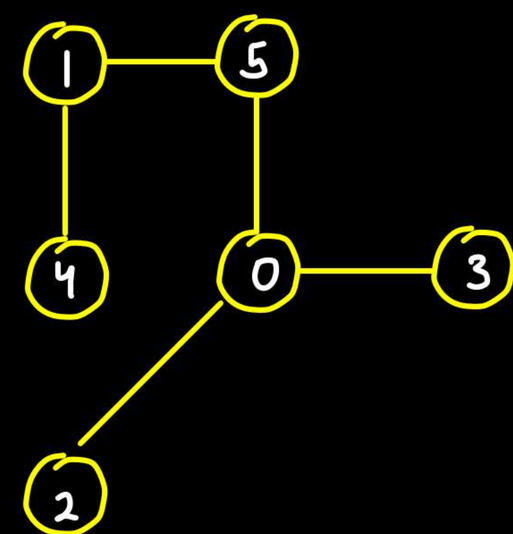
Adjacency List

LL → head (entire list)

Tree → Root (entire tree)

Graph → AL, AM



0 → {2, 3, 5}

1 → {4, 5}

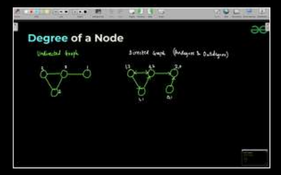
2 → {0}

3 → {0}

4 → {1}

5 → {0, 1}

edges = { {1,4}, {1,5}, {5,0}, {0,3}, {2,0} }



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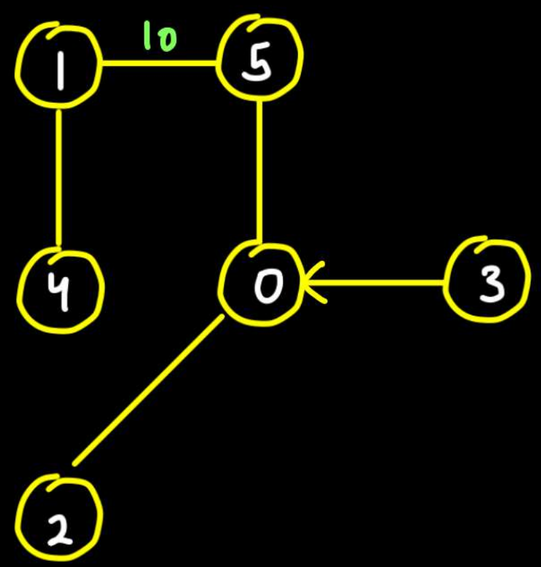
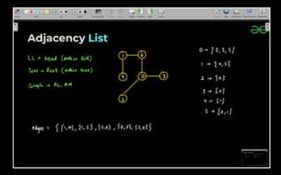
Adjacency Matrix



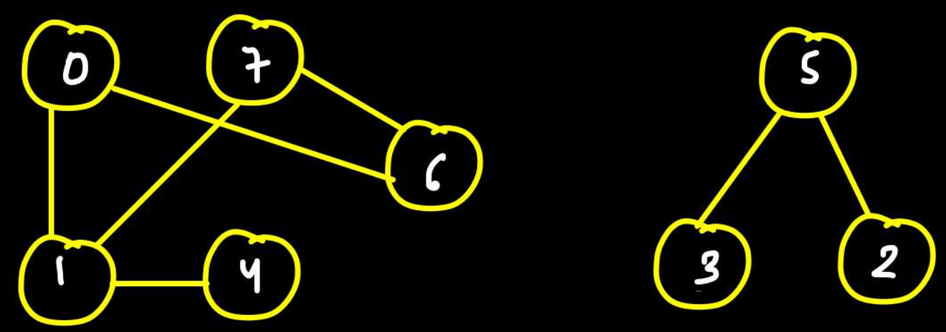
	0	1	2	3	4	5
0			✓			✓
1					✓	10
2	✓					
3	✓					
4		✓				
5	✓	✓				

Saves time (fetching)

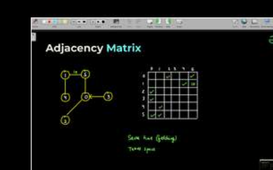
Takes space



Connected Components



This graph has 2 connected components



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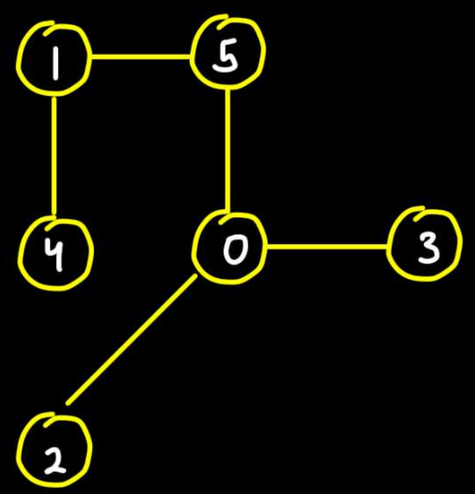
BFS and DFS (graph traversal)

BFS → Queue

DFS → Recursion

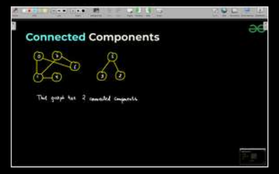


BFS Example

0 2 3 5 1 4

4
1
5
3
2
0

visited (set) or array

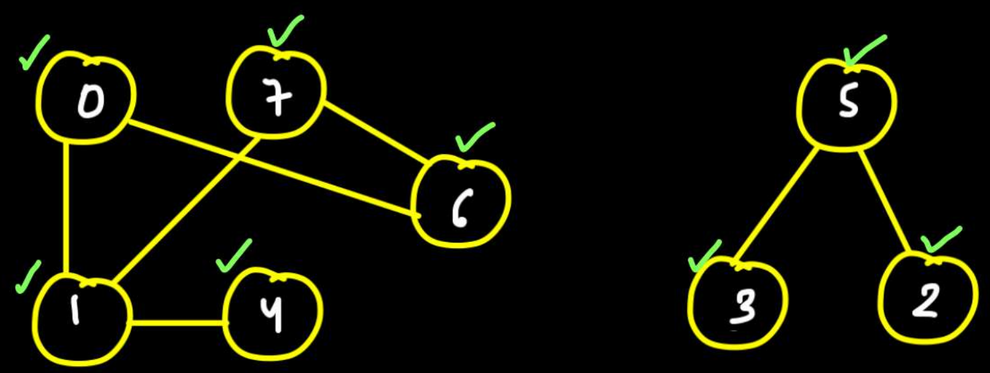


Apply BFS for all nodes (as starting pt.)

Ques: Number of Provinces (no. of components)

Connected components, BFS, AL, AM

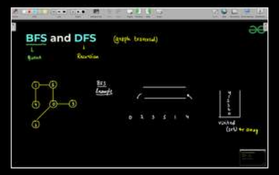
Count = ~~1~~ 2



V=8

vis

0	1	2	3	4	5	6	7
T	T	T	T	T	T	T	T



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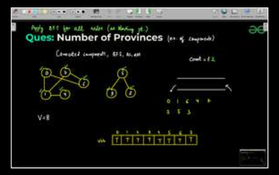
Ques: Keys and Rooms



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1 – land , 0 – water

Ques: Number of Islands

count = ~~0~~ ~~1~~ ~~2~~ 3

```
["1","1","1","1","0"],
["1","1","0","1","0"],
["1","1","0","0","0"],
["0","0","0","0","0"]
```

1

r-1, c
↑
r, c-1 ← r, c → r, c+1
↓
r+1, c

(0,0) (0,1) (1,0) (1,1)

(2,2) (3,3) (3,4)

	0	1	2	3	4
0	"1"	"1"	"0"	"0"	"0"
1	"1"	"1"	"0"	"0"	"0"
2	"0"	"0"	"1"	"0"	"0"
3	"0"	"0"	"0"	"1"	"1"

3

	0	1	2	3	4
0	T	T			
1	T	T			
2			T		
3				T	T

vis

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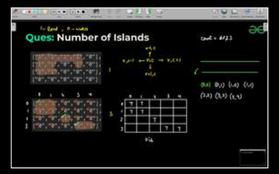
HW Find the number of islands (GFG Practice)



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Ques: Rotten Oranges

↓

Homework:

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THANKYOU

Cuties